CREDIT CARD APPROVAL PREDICTION

Project Documentation

Submitted as part of

APSCHE – SmartBridge AI & Machine Learning Internship

Project Title:

Credit Card Approval Prediction

Developed By

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1. Introduction

Credit card approval is a critical process in the banking and financial sector. Financial institutions receive numerous credit card applications every day and must evaluate each applicant's eligibility based on various financial and personal attributes. Traditional manual evaluation is often time-consuming and may lead to inconsistencies in decision-making.

The **Credit Card Approval Prediction** project applies Machine Learning techniques to automate the approval prediction process. By analyzing applicant information such as income, education, employment status, family details, and other relevant features, the system predicts whether a credit card application is likely to be approved or rejected.

The project follows a complete Machine Learning workflow, including data collection, preprocessing, feature encoding, model training, evaluation, and deployment. Multiple classification algorithms were trained and compared, with **Logistic Regression** achieving the highest prediction accuracy of **98.31%**, making it the final model for deployment.

To make the solution accessible to end users, a user-friendly web application was developed using the **Flask** framework. The application allows users to enter applicant details through a web interface and instantly receive the prediction result. The application has also been deployed on the **Render** cloud platform, enabling online access from any device with an internet connection.

This project demonstrates the practical application of Machine Learning, web development, and cloud deployment to build an intelligent decision-support system for credit card approval prediction.

2. Project Overview

The Credit Card Approval Prediction project is a Machine Learning-based web application developed to predict whether a credit card application is likely to be approved or rejected based on applicant information. The system assists financial institutions by providing a quick and data-driven prediction, reducing the need for manual evaluation.

The project utilizes a publicly available credit card application dataset containing demographic, financial, and employment-related information. The collected data underwent preprocessing steps such as handling missing values, encoding categorical variables, feature selection, and data transformation to prepare it for model training.

Several Machine Learning classification algorithms, including Logistic Regression, Decision Tree, Random Forest, and XGBoost, were trained and evaluated. After comparing their performance, Logistic Regression was selected as the final model because it achieved the highest prediction accuracy of **98.31%** on the test dataset.

A Flask-based web application was developed to provide a simple and interactive interface where users can enter applicant details and receive an instant prediction. The application was successfully deployed on the Render cloud platform, allowing users to access it through a web browser without requiring local installation.

The project integrates Machine Learning, web development, and cloud deployment into a single solution, demonstrating the practical implementation of Artificial Intelligence in financial decision support systems.

3. Problem Statement

Financial institutions receive a large number of credit card applications every day, making the approval process both time-consuming and challenging. Traditional manual evaluation methods require significant human effort and may lead to inconsistent decisions due to subjective judgment and increasing application volumes.

An efficient and reliable prediction system is needed to assist financial organizations in evaluating applicants based on historical data and predefined attributes. By leveraging Machine Learning techniques, the approval process can be automated, enabling faster decision-making while maintaining consistency and accuracy.

The objective of this project is to develop a Credit Card Approval Prediction system that analyzes applicant information such as gender, income, education, employment status, housing type, family details, and other relevant factors to predict whether a credit card application is likely to be approved or rejected. The proposed solution aims to improve the efficiency of the approval process and support better decision-making through data-driven predictions.

4. Objectives

The primary objectives of the Credit Card Approval Prediction project are:

- To develop a Machine Learning model capable of predicting whether a credit card application will be approved or rejected based on applicant information.
- To preprocess and transform raw data by handling missing values, encoding categorical features, and preparing the dataset for model training.
- To compare the performance of multiple Machine Learning classification algorithms and select the most accurate model for prediction.
- To design and develop a user-friendly web application using the Flask framework for real-time credit card approval prediction.
- To deploy the application on the Render cloud platform, enabling users to access the system through a web browser.
- To evaluate the application's performance and reliability through testing, including Apache JMeter load testing.

5. Project Workflow

The Credit Card Approval Prediction project follows a systematic Machine Learning workflow, starting from data collection and ending with deployment of the web application. Each stage contributes to building an accurate and reliable prediction system.

Step 1: Data Collection

The credit card application dataset was collected and imported into the development environment. It contains applicant demographic, financial, employment, and family-related information required for prediction.

Step 2: Data Preprocessing

The dataset was cleaned by handling missing values, removing unnecessary columns, encoding categorical variables using label encoding, and transforming the data into a suitable format for Machine Learning algorithms.

Step 3: Feature Selection

Relevant input features influencing credit card approval were selected, while the target variable was separated for model training and evaluation.

Step 4: Model Training

Multiple Machine Learning classification algorithms, including Logistic Regression, Decision Tree, Random Forest, and XGBoost, were trained using the prepared dataset.

Step 5: Model Evaluation

The trained models were evaluated using performance metrics such as accuracy. Logistic Regression achieved the highest accuracy of **98.31%** and was selected as the final prediction model.

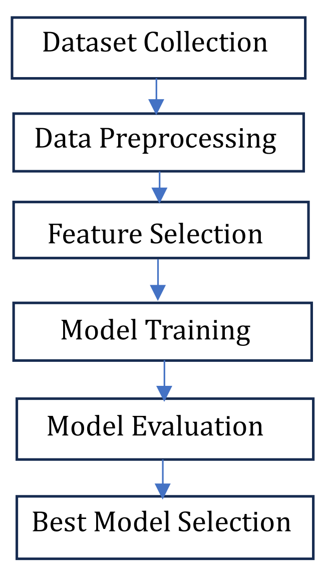
Step 6: Web Application Development

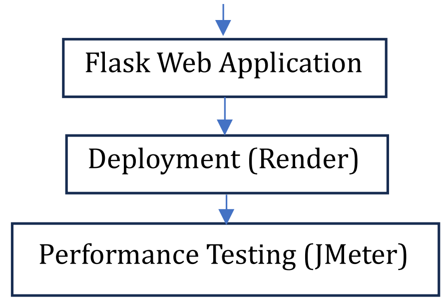
A Flask-based web application was developed to provide a simple interface where users can enter applicant details and receive real-time prediction results.

Step 7: Deployment and Testing

The completed application was deployed on the Render cloud platform for online accessibility. Performance testing was carried out using Apache JMeter to verify the application's reliability under multiple user requests.

Workflow Diagram





6. Dataset Description

The Credit Card Approval Prediction project uses a publicly available credit card application dataset consisting of applicant information and credit history records. The dataset was obtained for educational and research purposes to build a Machine Learning model capable of predicting whether a credit card application is likely to be approved or rejected.

The project utilizes two datasets:

- **application_record.csv** – Contains demographic, financial, employment, and family-related information of credit card applicants.
- **credit_record.csv** – Contains historical credit status records for each applicant.

These two datasets were combined and processed to generate the final dataset used for model training and prediction.

Dataset Features

The important attributes used in the project include:

- Gender
- Car Ownership
- Property Ownership
- Number of Children
- Annual Income
- Income Type
- Education Level
- Family Status
- Housing Type
- Mobile Phone Availability
- Work Phone Availability
- Phone Availability
- Email Availability
- Occupation Type
- Number of Family Members
- Age
- Years Employed

The target variable represents the credit card approval status, which is predicted using the trained Machine Learning model.

Before training, the dataset underwent preprocessing steps including handling missing values, encoding categorical variables, feature selection, and data transformation to improve model performance and ensure compatibility with Machine Learning algorithms.

7. Data Preprocessing

Data preprocessing is an essential step in the Machine Learning pipeline, as it improves data quality and prepares the dataset for model training. The raw datasets contained categorical attributes, missing values, and unnecessary information that required processing before building the prediction model.

The following preprocessing steps were performed:

7.1 Data Loading

The application_record.csv and credit_record.csv datasets were imported into the Python environment using the Pandas library and merged based on the common applicant ID.

7.2 Handling Missing Values

Missing values present in the dataset were identified and handled appropriately. Columns with excessive missing values were cleaned, and remaining missing entries were treated to ensure data consistency.

7.3 Data Cleaning

Duplicate records and unnecessary attributes that did not contribute to prediction were removed. The dataset was inspected to ensure that all records were valid and suitable for analysis.

7.4 Encoding Categorical Features

Machine Learning algorithms require numerical input. Therefore, categorical attributes such as Gender, Income Type, Education Type, Family Status, Housing Type, Occupation Type, Car Ownership, and Property Ownership were converted into numerical values using Label Encoding.

7.5 Feature Engineering

Additional features such as Age and Years Employed were derived from the original data to improve model performance and provide more meaningful input to the prediction model.

7.6 Feature Selection

The most relevant features influencing credit card approval were selected as input variables, while the approval status was chosen as the target variable for model training.

7.7 Train-Test Split

The processed dataset was divided into training and testing sets. The training dataset was used to train the Machine Learning models, while the testing dataset was used to evaluate their prediction performance.

After completing these preprocessing steps, the dataset became clean, consistent, and suitable for training multiple Machine Learning classification algorithms.

8. Model Building and Training

After completing data preprocessing, the prepared dataset was used to train multiple Machine Learning classification models. The objective was to identify the model that provides the highest prediction accuracy for credit card approval.

The dataset was divided into training and testing sets. The training dataset was used to build the models, while the testing dataset was used to evaluate their performance on unseen data.

The following Machine Learning algorithms were implemented and compared:

8.1 Logistic Regression

Logistic Regression is a supervised classification algorithm widely used for binary classification problems. It was trained using the processed dataset to predict whether a credit card application would be approved or rejected.

8.2 Decision Tree Classifier

The Decision Tree model classifies data by creating a tree-like structure of decision rules based on input features. It provides an interpretable approach for classification.

8.3 Random Forest Classifier

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

8.4 XGBoost Classifier

XGBoost is a gradient boosting algorithm that builds multiple decision trees sequentially to optimize prediction performance. It is known for its efficiency and high accuracy on structured datasets.

Each model was trained using the same training dataset and evaluated on the testing dataset. Their performance was compared using prediction accuracy, allowing the selection of the most suitable model for deployment in the Flask web application.

9. Best Model Selection

Several Machine Learning classification algorithms were trained and evaluated to identify the most suitable model for predicting credit card approval. Each model was tested using the same preprocessed dataset, and its performance was measured using prediction accuracy.

The evaluated models included:

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier
- XGBoost Classifier

After comparing the performance of all models, **Logistic Regression** achieved the highest prediction accuracy of **98.31%**, making it the best-performing model for this project.

Best Model Performance

Machine Learning Model	Performance
Logistic regression	98.31% Accuracy (Best Model)
Decision Tree	Evaluated
Random Forest	Evaluated
XGBoost	Evaluated

The Logistic Regression model was selected for deployment because it provided the highest prediction accuracy while maintaining fast prediction speed and reliable performance on the testing dataset.

The trained Logistic Regression model was saved as **best_model.pkl** using the Pickle library and later integrated into the Flask web application. During prediction, the application loads this model and generates real-time credit card approval predictions based on user-provided input data.

The model selection process demonstrates that evaluating multiple algorithms before deployment helps in choosing the most accurate and efficient solution for the given problem.

10. Flask Web Application Development

After selecting the Logistic Regression model as the best-performing classifier, a web application was developed using the **Flask** framework to provide an interactive interface for users. The Flask application enables users to enter applicant details through a web form and receive an instant prediction regarding credit card approval.

The application was developed using Python and follows a modular structure consisting of backend logic, HTML templates, CSS styling, the trained Machine Learning model, and supporting files.

Application Workflow

The workflow of the Flask application is as follows:

1. The user opens the home page of the application.
2. The user navigates to the prediction page and enters the required applicant details.
3. The submitted information is received by the Flask backend through an HTTP POST request.
4. Categorical input values are converted into numerical values using the saved label encoders.
5. The processed input data is passed to the trained Logistic Regression model.
6. The model predicts whether the credit card application is approved or rejected.
7. The prediction result is displayed to the user on the result page.

Backend Implementation

The Flask application is implemented in the app.py file. It loads the trained Machine Learning model (best_model.pkl) and the saved label encoders (label_encoders.pkl) during application startup. Flask routes handle navigation between different web pages and process user input for prediction.

Frontend Interface

The user interface was developed using **HTML** and **CSS**. The application consists of the following pages:

- **Home Page** – Introduces the application and provides navigation to the prediction page.
- **Prediction Page** – Allows users to enter applicant details required for prediction.
- **Result Page** – Displays the prediction result as either **Credit Card Approved** or **Credit Card Rejected**.

The integration of Flask with the trained Machine Learning model enables real-time predictions through a simple, user-friendly web interface, making the application suitable for demonstration and practical use.

11.Frontend Development

A simple and user-friendly frontend was developed to enable users to interact with the Credit Card Approval Prediction system. The frontend was built using **HTML** for the page structure and

CSS for styling, while the Flask framework was used to connect the frontend with the Machine Learning backend.

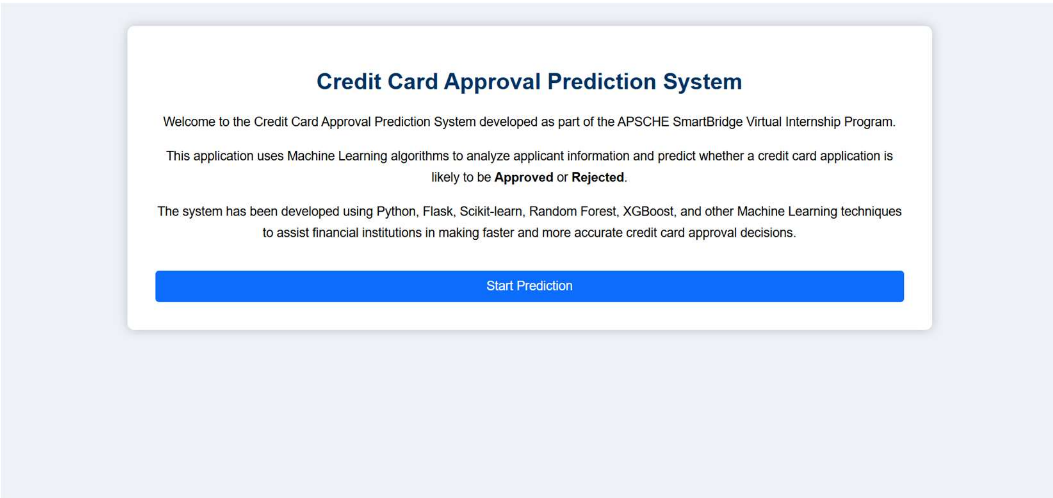
The application consists of three main web pages that provide a smooth and intuitive user experience.

11.1 Home Page

The Home Page serves as the entry point of the application. It provides a brief introduction to the Credit Card Approval Prediction system and contains a navigation button that directs users to the prediction page.

Features:

- Project title and description
- Clean and responsive interface
- Navigation to the prediction page



11.2 Prediction Page

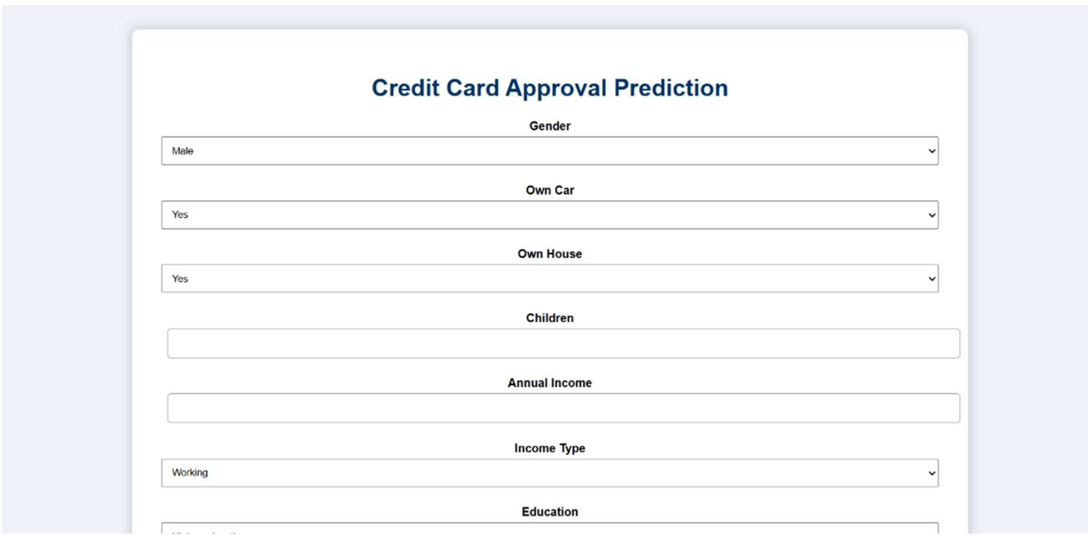
The Prediction Page contains an input form where users enter applicant details required for prediction. The form collects demographic, financial, employment, and family-related information, which is submitted to the Flask backend for processing.

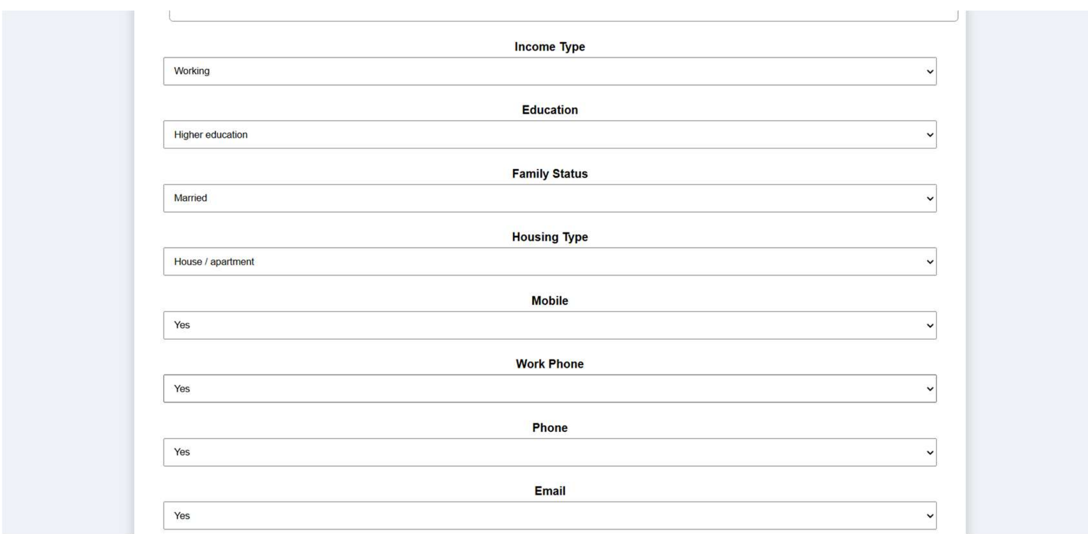
Input Fields Include:

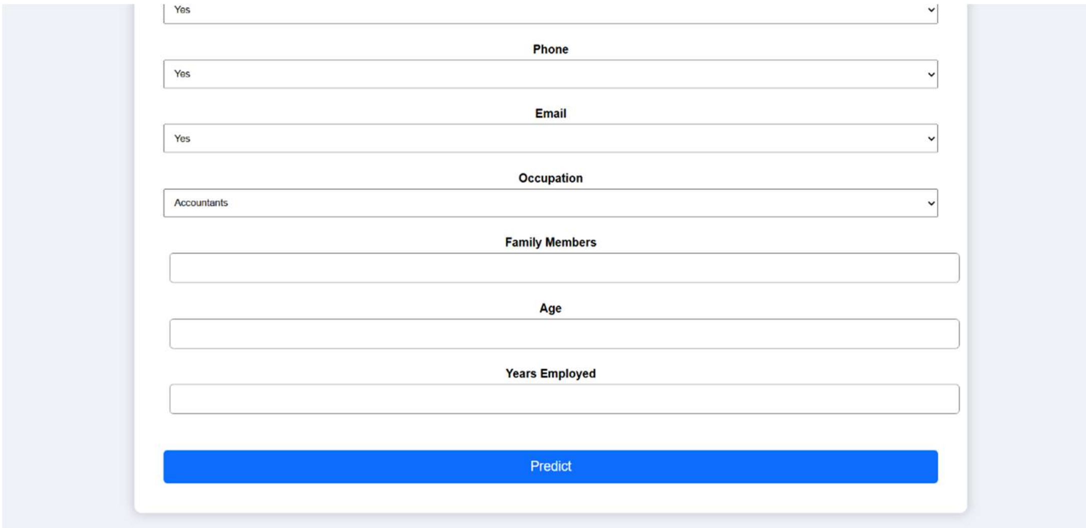
- Gender
- Car Ownership
- Property Ownership
- Number of Children
- Annual Income
- Income Type
- Education Type

- Family Status
- Housing Type
- Mobile, Work Phone, Phone, and Email Availability
- Occupation Type
- Number of Family Members
- Age
- Years Employed

After entering the required information, users click the **Predict** button to generate the prediction.





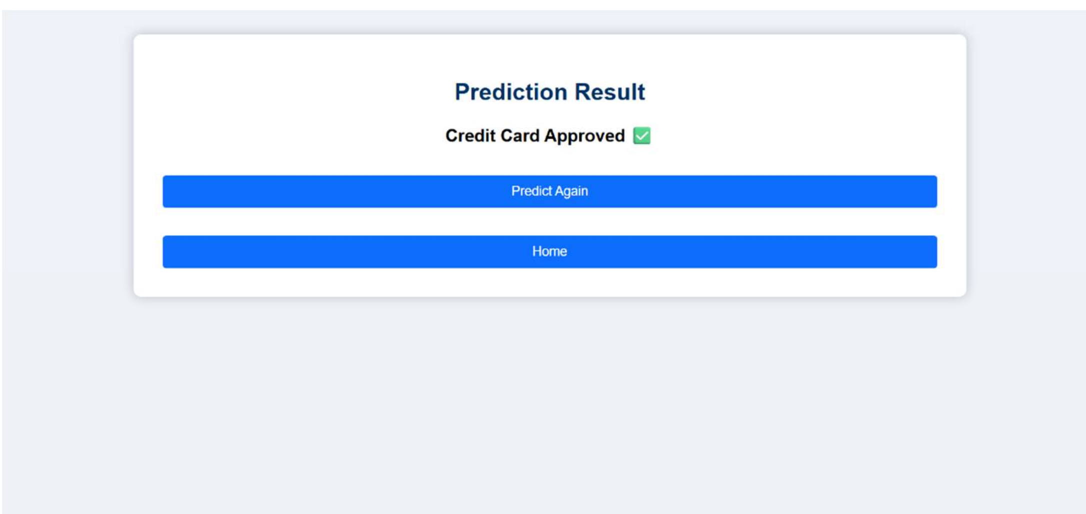


11.3 Result Page

The Result Page displays the prediction generated by the Machine Learning model. Based on the applicant's details, the system displays one of the following results:

- **Credit Card Approved** ✓
- **Credit Card Rejected** ✕

The result page provides immediate feedback, allowing users to quickly understand the model's prediction.



Frontend Summary

The frontend was designed with simplicity, responsiveness, and ease of use in mind. By combining HTML, CSS, and Flask templates, the application provides an efficient interface for collecting user inputs and displaying prediction results in real time.

12.Application Deployment using Render

After successful development and testing, the Credit Card Approval Prediction web application was deployed on the Render cloud platform. Deployment enables users to access the application through a web browser without installing any software or configuring a local environment.

Render was selected because it provides a simple deployment process for Flask applications, supports GitHub integration, and offers free hosting for educational and demonstration purposes.

Deployment Process

The deployment was completed using the following steps:

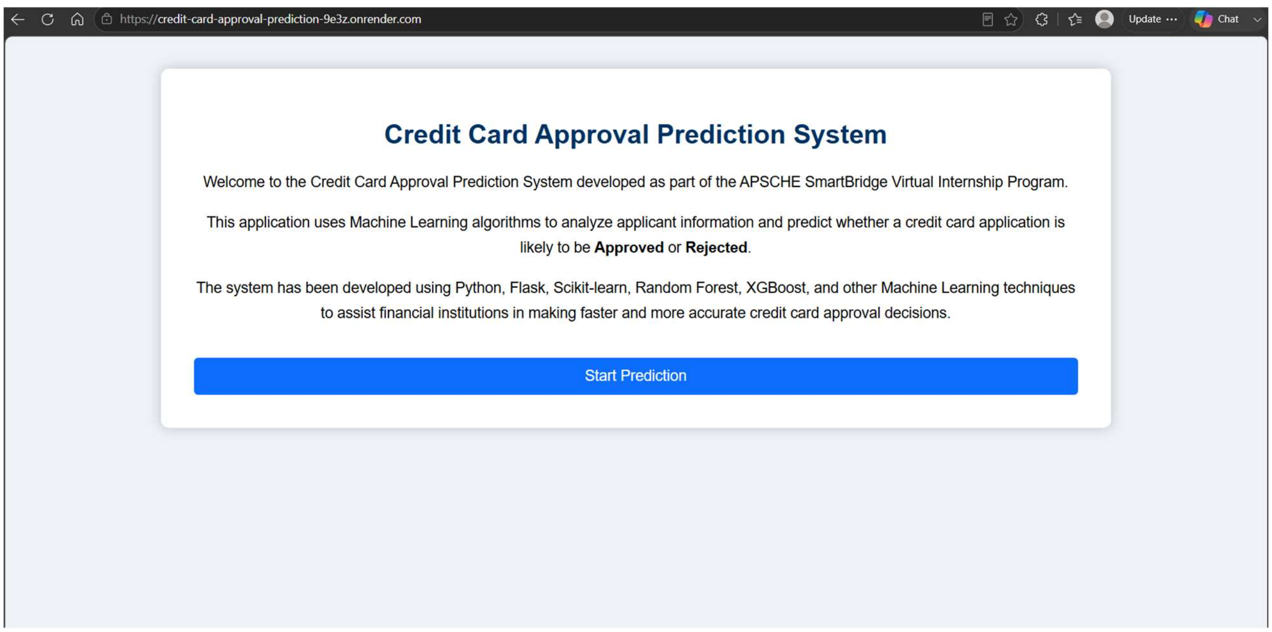
1. The complete project was uploaded to a GitHub repository.
2. A new Web Service was created on the Render platform.
3. The GitHub repository was connected to the Render service.
4. Required dependencies were installed automatically using the **requirements.txt** file.
5. The Flask application was started using the **Procfile** configuration.
6. After successful deployment, the application became accessible through a public URL.

Deployed Application

Deployment Platform: Render

Application URL:

https://credit-card-approval-prediction-9e3z.onrender.com



Benefits of Deployment

- The application can be accessed from any device with an internet connection.
- No local installation is required for end users.
- Easy integration with GitHub for automatic deployment updates.
- Demonstrates practical cloud deployment of a Machine Learning web application.
- Simplifies project sharing, evaluation, and demonstration.

13.Performance Testing Using Apache JMeter

To evaluate the performance and reliability of the Credit Card Approval Prediction web application, **Apache JMeter** was used for load testing. The objective was to analyze the application's behaviour when multiple users accessed it simultaneously and to verify that it responded correctly under load.

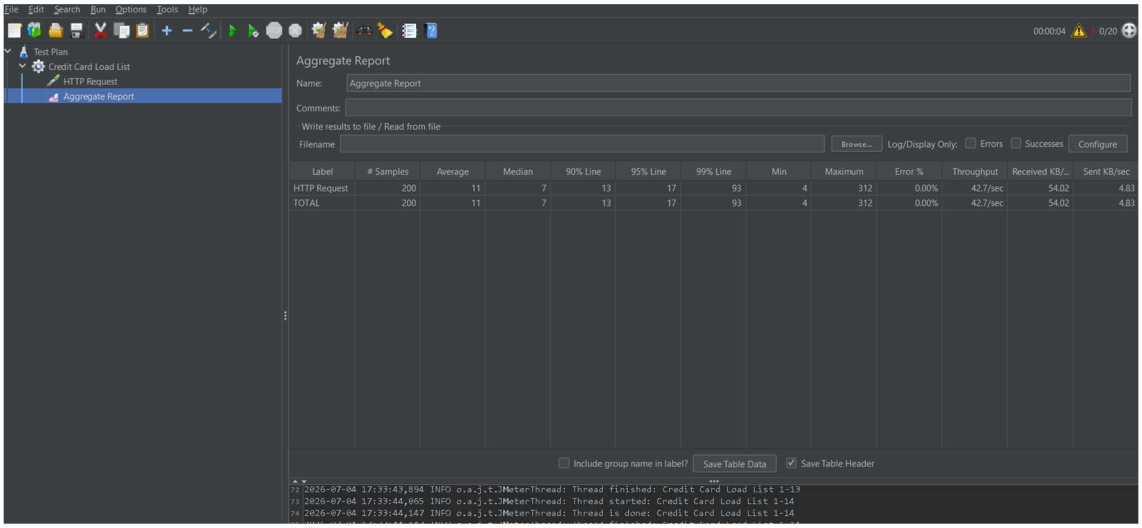
Test Configuration

The following test configuration was used during performance testing:

Parameter	Value
Testing Tool	Apache JMeter 5.6.3
Thread Group Name	Credit Card Load Test
Number of Threads (Users)	20
Ramp-up Period	5 seconds
Loop Count	10
Protocol	HTTP
Server	127.0.0.1
Port Number	5000
Request Method	GET
Request Path	/

Aggregate Report

The performance test results were analyzed using the Aggregate Report listener in Apache JMeter.



Test Results

The Aggregate Report provides important performance metrics such as:

- Number of samples executed
- Average response time
- Minimum and maximum response time
- Standard deviation
- Error percentage
- Throughput
- Network received and sent per second

The obtained results indicated that the application successfully handled multiple concurrent requests with **0% errors**, demonstrating stable performance during testing.

Observations

- The Flask application successfully processed all incoming requests.
- No request failures or server errors were observed during the load test.
- Response times remained stable throughout the execution.
- The application performed reliably under the configured user load.

Conclusion

The Apache JMeter performance test confirms that the Credit Card Approval Prediction application is capable of handling multiple simultaneous users efficiently. The results demonstrate that the application is stable, responsive, and suitable for deployment and demonstration purposes.

14. Results and Discussion

The Credit Card Approval Prediction project was successfully completed by implementing a complete Machine Learning pipeline, developing a Flask-based web application, deploying the application on the Render cloud platform, and validating its performance through load testing.

Machine Learning Results

Multiple Machine Learning classification algorithms were trained and evaluated using the prepared dataset. After comparing their performance, **Logistic Regression** achieved the highest prediction accuracy of **98.31%** and was selected as the final model for deployment.

The trained model demonstrated excellent prediction capability, producing reliable results for unseen test data while maintaining efficient execution time.

Web Application Results

The Flask web application was successfully integrated with the trained Machine Learning model. Users can enter applicant details through a simple web interface and receive instant credit card approval predictions.

The application correctly performs the following tasks:

- Accepts applicant information from users.
- Processes categorical and numerical input values.
- Applies label encoding to categorical features.
- Generates predictions using the trained Logistic Regression model.
- Displays the prediction result on the result page.

Deployment Results

The application was successfully deployed on the **Render** cloud platform, allowing users to access the system online without requiring local installation. The deployed application functioned correctly and provided prediction results through a web browser.

Performance Testing Results

Performance testing using **Apache JMeter** demonstrated that the application could process multiple simultaneous requests successfully. The Aggregate Report showed stable response times and **0% error rate**, indicating reliable application performance under the configured load.

Overall Discussion

The project successfully combines Machine Learning, Flask web development, and cloud deployment into a practical credit card approval prediction system. The high prediction accuracy, user-friendly interface, successful cloud deployment, and satisfactory performance testing results demonstrate that the developed solution is accurate, reliable, and suitable for educational and demonstration purposes.

15.Challenges Faced

During the development of the Credit Card Approval Prediction project, several technical and implementation challenges were encountered. These challenges were resolved through systematic debugging, testing, and collaboration among team members.

Challenge 1: Data Preprocessing

The datasets contained missing values and categorical attributes that required preprocessing before model training.

Solution: Missing values were handled appropriately, and categorical features were converted into numerical values using Label Encoding.

Challenge 2: Model Selection

Different Machine Learning algorithms produced varying prediction accuracies, making it necessary to identify the most suitable model.

Solution: Logistic Regression, Decision Tree, Random Forest, and XGBoost models were trained and evaluated. Logistic Regression achieved the highest accuracy (98.31%) and was selected as the final model.

Challenge 3: Flask Integration

Integrating the trained Machine Learning model with the Flask web application required proper handling of user input and preprocessing.

Solution: The trained model and label encoders were saved using Pickle and successfully integrated into the Flask backend for real-time prediction.

Challenge 4: Cloud Deployment

Deploying the application on the Render platform required proper project configuration, including dependency management and application startup settings.

Solution: The project was configured using the requirements.txt and Procfile files, enabling successful deployment on Render.

Challenge 5: Performance Testing

Ensuring that the application remained responsive under multiple simultaneous user requests was an important requirement.

Solution: Apache JMeter was used to perform load testing. The application successfully handled the configured user load with stable performance and no request failures.

16.Future Scope

Although the current Credit Card Approval Prediction system performs efficiently, several enhancements can be incorporated to improve its functionality, scalability, and user experience in the future.

The following improvements are proposed:

- Integrate larger and more diverse datasets to improve the model's prediction accuracy and generalization.
- Implement user authentication and role-based access control to enhance application security.
- Store applicant information and prediction history using a relational database such as MySQL or PostgreSQL.
- Develop an interactive dashboard for visualizing prediction statistics and application analytics.
- Deploy the application on a scalable cloud platform with automatic scaling and monitoring capabilities.
- Explore advanced Machine Learning and Deep Learning algorithms to further improve prediction performance.
- Develop a mobile-friendly interface or Android application for easier accessibility.

These enhancements will improve the application's usability, security, scalability, and suitability for real-world financial environments.

17.Conclusion

The Credit Card Approval Prediction project successfully demonstrates the practical application of Machine Learning in automating credit card approval decisions. The project involved data preprocessing, feature engineering, model training, evaluation, web application development, cloud deployment, and performance testing.

Several Machine Learning classification algorithms were evaluated, and **Logistic Regression** achieved the highest prediction accuracy of **98.31%**, making it the most suitable model for deployment. The trained model was successfully integrated into a Flask web application, enabling users to obtain instant credit card approval predictions through a simple and user-friendly interface.

The application was deployed on the Render cloud platform, allowing online access, and its performance was validated using Apache JMeter, where it demonstrated stable operation under multiple concurrent requests.

Overall, the project successfully integrates Machine Learning, web development, and cloud deployment into a complete prediction system. It provides an effective solution for assisting financial institutions in making faster, more consistent, and data-driven credit card approval decisions while serving as a valuable learning experience in developing end-to-end Artificial Intelligence applications.

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